

Tyler Graff

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Technical Fellow | Governed AI, Cyber Analytics, and Distributed Systems

DoD TS/SCI CI Poly | Security+ | B.S. & M.S. Electrical Engineering

SUMMARY

Hands-on customer-facing Technical Fellow and systems architect with 20+ years of experience building fieldable systems across cyber, distributed systems, embedded systems, defense technology, and artificial intelligence disciplines.

Current work centers on architecting governed AI platforms for defense use cases, including state-machine-constrained LLM workflows, scoped tool execution, deterministic testing, comprehensive audit logging, and air-gapped deployment models.

Past work includes high-throughput cyber analytics, 10Gbps packet-processing systems, embedded hardware, RF prototyping, and reverse engineering.

CORE STRENGTHS

- Governed AI platforms, constrained LLM workflows, auditability, and secure deployment models
- Cyber analytics, endpoint telemetry, packet processing, and high-throughput data systems
- Distributed systems, custom storage, ETL pipelines, APIs, and performance-oriented infrastructure
- Embedded systems, RF prototyping, kernel-level software, and hardware/software integration
- Defense/federal solution architecture, RFIs, white papers, proposals, customer discovery, and technical capture
- Technical leadership from R&D concept through fieldable deployment

EXPERIENCE

MetroStar Systems, Inc.

Technical Fellow and AI Product Owner *January 2025 - Present*

Led architecture and product ownership for Midas, MetroStar's governed AI platform for structured, repeatable, auditable workflows in defense and federal environments.

- Designed AI solution architecture across CDAO JATIC, CMPMI, and internal initiatives; developed the technical solution approach for C2SET and supported proposal authorship via the GCCH Enclave.
- Led 7 FTE team to transform a prototype AI system into a modular workflow platform resulting in \$2M in business across multiple pilot customers.
- Designed core architecture: plugin-based internal system bus, state-machine LLM orchestration, scoped agent/tool execution, deterministic test harness, and audit-oriented execution model.
- Defined reusable workflow patterns for RMF-aligned assessment, continuous ATO support, software artifact review, and controlled technical-document evaluation.
- Architected deployment model for secure IL5 and air-gapped environments.

- Established gold-file/snapshot testing strategy to enforce deterministic behavior, expose regressions, and document platform behavior.
- Led products to awardable status on Tradewinds Solutions Marketplace and DoD Platform One Marketplace.

Senior Software Engineer / Technical Lead, CDAO JATIC Program *January 2024 - January 2025*

Owned technical roadmap and execution for a 10+ engineer effort to adapt, deploy, and maintain the open-source Nebari platform for CDAO, including ATO preparation at DoD IL5.

- Led Program Increment and Sprint planning; directed the technical effort to deploy the platform into a cloud-based air-gapped enclave.
- Led software security and ATO effort: defined and executed tasking to generate SBOM, identify and resolve CVE findings, and produce functional diagrams describing the security surface area.
- Enabled promotion of team members from outside collaborators to full members of the upstream open-source community.

Voreas Laboratories, Inc.

Director / Engineering Manager, Systems Engineering *February 2018 - February 2023*

Built and led engineering teams responsible for a custom cybersecurity analytics platform across AWS, Azure, and on-prem customer deployments.

- Established and led a 5-FTE startup engineering team building a cloud-agnostic distributed analytics platform in C, Ruby, and Python.
- Delivered and deployed the inaugural customer platform on time and on budget for integration with real-time customer data pipelines.
- Supported \$3M+ in profitable revenue by delivering 10+ targeted analytics capabilities that materially increased customer coverage, throughput, and analytic performance.
- Built high-throughput infrastructure to ingest 10+ TB/day of heterogeneous endpoint telemetry into a purpose-built distributed key-value store.
- Enabled correlation of temporally separated event UUIDs into linear process histories for malware and CNO analysis.

Pindrop Security, Inc.

Engineering Manager, Software Tooling *July 2016 - February 2018*

- Founded and led a 6-FTE tooling team supporting Pindrop's flagship fraud-detection platform.
- Led custom 10Gbps VoIP traffic-simulation effort to quantify performance risks before customer deployment.
- Reduced release and deployment risk by executing build-system redesign that increased build frequency from quarterly to daily.
- Automated customer-facing demo systems in partnership with the CEO and Sales organization, reducing demo lead time from days to hours.
- Built internal tooling in C/C++, Ruby, Python, and Go.

Damballa, Inc.

Staff Engineer, Sensor Team *February 2014 - June 2016*

- Recruited to increase cybersecurity platform traffic inspection capability from 1Gbps to 10Gbps.
- Led a 3-FTE team to implement and deploy 10Gbps inspection capability on time.

- Implemented zero-copy Linux kernel module tightly coupled to custom deep packet inspection architecture.
 - Achieved required 10Gbps capability while reducing hardware footprint and supporting customer renewals.
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EARLIER EXPERIENCE

- **MapLarge, Inc. - Director, Program Management, 2023:** Led customer-facing maintenance and upgrade of a scalable geospatial data platform in an operational DoD environment; aligned custom analytics work with reusable platform capabilities under staffing and budget constraints.
 - **Prolaag LLC - Director, Hardware, Part-Time, 2013-2019:** Designed custom low-power security hardware for a Layer 3 VPN device, including multiprocessor microcontroller architecture, electrical/mechanical/thermal design, and design-for-manufacturing. Developed avalanche-breakdown true-random-number-generator circuitry and signal-processing algorithms that passed Diehard statistical testing without software post-processing.
 - **Global InfoTek, Inc. - Senior Engineer, 2012-2014:** Led Windows-internals reverse-engineering and development in C, C++, and Python; developed Win7 x64 kernel driver routines and system-call hooks; led a 3-FTE team to deliver a \$1M contract.
 - **Ball Aerospace, Inc. - Software Engineer, 2009-2012:** Supported prototype airborne IR LIDAR development and field operation; built operator consoles and test systems; led spacecraft telemetry analysis/reporting software; supported field testing and on-orbit commissioning telemetry.
 - **Cellnet Technology, Inc. - Firmware Engineering Intern, 2007-2008:** Prototyped low-cost, low-power spread-spectrum RF hardware and developed a programming adapter for legacy deployed hardware. Awarded US Patent #8,111,063 (Magnetic Rotation Sensing) for an electro-mechanical hardware implementation and related software algorithm to detect and minimize hardware failures over device lifetime.
 - **National Instruments - Firmware Development Engineering Intern, 2006:** Developed firmware in C and VHDL for an embedded USB data acquisition device; built a VHDL test bench achieving 100% code coverage and verification of external signal timing; served as a technical representative for Georgia Tech.
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CREDENTIALS

- **M.S., Electrical Engineering** - University of Colorado, 2009
- **B.S., Electrical Engineering** - Georgia Institute of Technology, 2006
- **Ontologize: Palantir AIP & Foundry Foundations** - December 2025
- **CompTIA Security+** - May 2024
- **FCC Amateur Extra Class License** - 2006
- **US Patent #8,111,063** - Magnetic Rotation Sensing, 2012